Quantitative Image Analysis of Bismuth Electrodeposition on Carbon Fiber Networks

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Objective and contribution

Carbon-fiber texture complicates interpretation of deposition-related image changes. This project evaluated whether grayscale features distinguish pristine and bismuth (Bi)-deposited Toray carbon paper or track deposition potential and measured Bi loading, then explored spatial candidate measurement. I constructed cohorts, designed specimen-level inference, and implemented configurable Python workflows for statistical analysis and exploratory segmentation.

Data and statistical design

Across four magnifications (65×–260×), 116 images represented 20 Bi electrodes and five pristine specimens. Repeated views were aggregated within specimen and magnification to avoid pseudoreplication; magnifications were analyzed separately because their intensity distributions differed. Images shared a fixed 0–1 grayscale scale. Two-sided label permutation tests assessed group differences. Jensen–Shannon histogram PERMANOVA with dispersion checks tested potential effects beyond scalar metrics; potential-adjusted histogram PERMANOVA and scalar permutation ANCOVA assessed loading while accounting for deposition potential (17 electrodes; 16 at 260×). Holm correction covered four magnifications within each endpoint or model term.

Whole image findings

Mean intensity was higher in the Bi group, reaching significance after correction only at 260× (difference $= 0.0207$; adjusted $p = 0.0024$). $P_{95} - \text{median}$ decreased significantly at all four magnifications, indicating a smaller gap between median and 95th-percentile intensities (Figure 1). These patterns describe a redistribution of grayscale intensities. Potential effects and potential-adjusted loading associations remained inconclusive after correction; intensity was not established as a loading proxy.

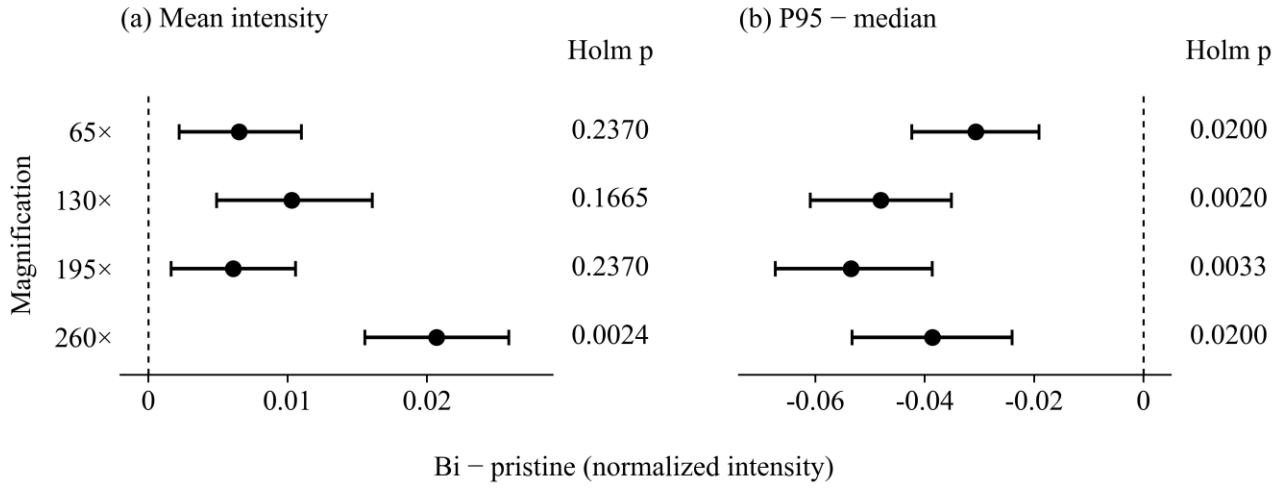


Figure 1. Bi-minus-pristine differences on a fixed 0–1 grayscale scale. Points show group-mean differences; bars, unadjusted 95% bootstrap confidence intervals; labels, Holm-adjusted permutation p-values. Pristine n = 5; pooled Bi n = 20 (19 at 260×). Inference follows the adjusted tests.

Spatial analysis and method selection

Whole-image summaries do not retain feature locations or identify individual structures, motivating spatial analysis within the carbon-fiber background. Initial thresholding, ridge-filter, and edge comparisons at 260× did not yield a complete carbon-fiber mask. I therefore used masks to define support regions around locally bright candidates. Local-z seeding measures brightness relative to the neighborhood; grouping nearby cores controls marker initialization. Entire connected Sauvola-mask regions overlapping seeds were retained. Regions with multiple marker groups were partitioned by watershed on their negative Euclidean distance transforms, preserving the total retained area. These exploratory choices do not establish Bi-specific boundaries.

Worked example and outputs

One 160 mV, 260× image yielded 190 candidate labels from nine support regions; 59 labels had at least one quality control (QC) flag (Figure 2). Segment, image, and electrode tables report counts, areas, equivalent-circle diameters, and flags for small, boundary-touching, or disconnected labels. Flags do not automatically exclude candidates.

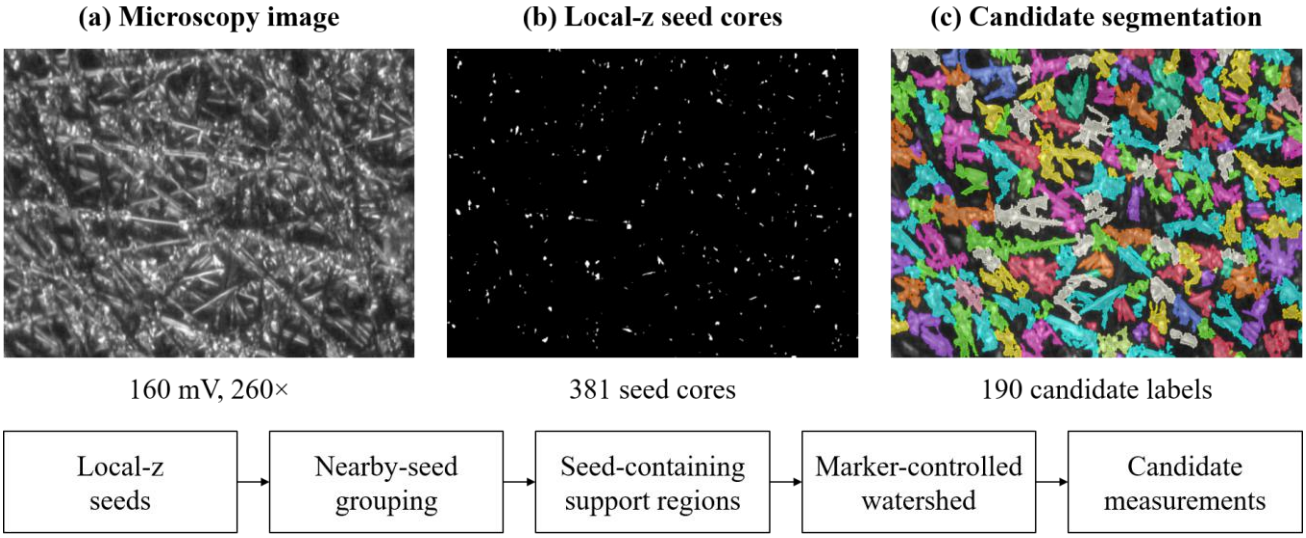


Figure 2. Worked 260× image: original, local-z seed cores, and candidate labels, with workflow below. Colors distinguish labels.

Implementation and further validation

Parameter sweeps and overlays informed exploratory choices of neighborhood size and seed-merging gap. Saved settings and run records trace inputs and outputs. I separated measurement and morphology QC from segmentation so saved labels can be remeasured without repeating detection, linking image features to inspectable candidate-level measurements.

Parameters remain illustrative. Segmentation accuracy and candidate Bi identity require reference annotations, independent material confirmation, and sensitivity analysis. Areas and diameters remain in pixels^2 and pixels until spatial calibration.